

STEAMROOT · DESIGN DECISION RECORD

SteamRoot recommendations accepted

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DOCUMENT CONTROL

Document ID	STR-DDR-002	Version	0.3
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/steamroot		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Item 1 consequence updated: evaporator bank now in the model within R7 (STR-CAL-001 v0.4)
0.2	2026-10-02	Amish Chadha	Items 10 to 12 decided by Amish on 2026-10-02 (STR-DEC-001, items 1 to 3); the ruling is still to be obtained
0.1	2026-09-25	Amish Chadha	Record the TRL 3 review recommendations accepted by Amish, what changed in the repo, and the items still open

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 1 to 9); items 10 to 12 remained open at this record and were decided by Amish on 2026-10-02 (STR-DEC-001, items 1 to 3): "i approve your recommendations for all 555 open decisions." The ruling of item 10 is still to be obtained.

■ Context

The TRL 3 review note (docs/REVIEW.md , session 2026-09-25) and STR-DDR-001 left several items "Proposed, awaiting Amish", most with a recommendation. On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Every open item that carried a recommendation is therefore decided as recommended. Where the recommendation listed several options, the recommended option is the decision. Items with no recommendation stay open. TRL 4 remains on hold by Amish's earlier instruction, and the repo stays at `trl: 3 , trl_target: 3`.

Safety: None of these decisions replaces the written ruling from the local boiler authority, which is still an open prerequisite for any detail design or build.

■ Decision

Table 1. Items decided by Amish on 2026-09-25.

#	Item	Decision	What changed in the repo
1	R5 efficiency and R4 wood use	Decided by Amish, 2026-09-25: go with recommendation. Keep the targets (65 %, 4 kg/m ²) and study a convective evaporator bank in the flue with controlled primary and secondary air at the next paper step.	STR-CAL-001 v0.2, section 4, adds the study: about 3.3 m of 25.4 mm tube (0.26 m ²) reaches 65 % at $\lambda = 2.0$; almost none is needed at $\lambda = 1.5$. R4 needs about 72 %. Baseline design unchanged; R4 and R5 still not met.
2	R2 treatment rate	Decided by Amish, 2026-09-25: go with recommendation (a). Accept about 1.9 m ² /h at 15 cm and restate the 5 cm target at about 3.4 m ² /h, until soil tests exist.	STR-REQ-001 v0.4: R2 from 2 and 5 m ² /h to 1.85 (about 1.9) and 3.4 m ² /h. Status from not met to at risk (1.88 and 3.41 m ² /h).
3	R6 pressure	Decided by Amish, 2026-09-25: go with recommendation. 0.1 bar gauge at the header, where the vent limit applies, and 0.15 bar gauge at the coil inlet.	STR-REQ-001 v0.4: R6 split. Status from not met (0.110 bar against 0.1) to met (header 0.033, inlet 0.110). Drawing note updated.
4	R9 relief valve	Decided by Amish, 2026-09-25: go with recommendation (both parts). Word the set pressure as "15 psi (1.03 bar) or less" and size the valve for the dry-coil refeed flash (about 155 kg/h, 14 mm orifice) as well as 30 kg/h.	STR-REQ-001 v0.4 reworded. Maker's capacity table checked: Watts Series 315, 3/4 in, 15 psi, 375 lb/h (170 kg/h), so the BOM valve is kept. Status from at risk to met on paper. BOM note, drawing note and CAL-001 updated.
5	R11 mass	Decided by Amish, 2026-09-25: go with recommendation (a). "500 kg or less as towed, with the tank drained"; fill on site.	STR-REQ-001 v0.4 restated. Status from not met (536 kg full) to met (402 kg as towed). Width stays at risk (1.48 m).
6	R12 budget	Decided by Amish, 2026-09-25: go with recommendation (a). Raise the budget to about \$2,200 for about 15 % contingency.	<code>project.yaml</code> <code>budget_usd</code> from 1800 to 2200; R12 from \$1,800 to \$2,200. Status from not met to met (\$1,915, \$285 contingency).
7	Firebox lining	Decided by Amish, 2026-09-25: go with recommendation. Ceramic fiber lining, not castable (castable would add about 200 kg).	Recorded in STR-PRC-001 v0.4. Design already used fiber; no geometry change.
8	Hose between hoods	Decided by Amish, 2026-09-25: go with recommendation. One hose moved between hoods with a steam-rated coupling, after steam is diverted to the vent.	Recorded in STR-PRC-001 v0.4. BOM already had one hose; no change.
9	Fuel	Decided by Amish, 2026-09-25: go with recommendation (the design proposal). Wood only, no liquid fuel backup.	Recorded in STR-PRC-001 v0.4 and STR-DDR-001 v0.2.

■ Items still open

Table 2. Items left open at this record; decided on 2026-10-02.

#	Item	Status
10	Written ruling from the local boiler authority	Approved as a prerequisite; not obtained. Blocks detail design and any build. Confirmed by Amish on 2026-10-02 (STR-DEC-001, item 1): the design is reviewed against the ruling and anything stricter is adopted.
11	Which jurisdiction is "home" and which authority to ask	Decided by Amish, 2026-10-02 (STR-DEC-001, item 2): Texas; the ruling is asked of the Texas Department of Licensing and Regulation's boiler program.
12	The named market garden and nursery (first co-design partners)	Decided by Amish, 2026-10-02 (STR-DEC-001, item 3): one market garden and one nursery within driving distance of Irving that already steam, solarize or chemically treat soil, found through Texas A&M AgriLife Extension; none is named or agreed yet.

■ Consequences

- Requirement status (STR-CAL-001 v0.2, 15 checks): 2 not met (R4, R5), 3 at risk (R2 at 15 cm and at 5 cm, R11 width), 1 not verifiable at TRL 3 (R1), 9 met. Before these decisions: 7 not met, 2 at risk, 1 not verifiable, 4 met.
- Documents bumped: STR-PRB-001 v0.4, STR-PRC-001 v0.4, STR-REQ-001 v0.4, STR-CAL-001 v0.2, STR-DDR-001 v0.2. Drawing STR-DWG-002 moved to Rev P2 for note changes (relief valve rating, towed mass, R6 limits); geometry is unchanged.
- The evaporator bank was added to the model on 2026-10-02: 6.4 m of 15.88 mm tube with one coil turn fewer keeps the heated section at 7.3 L, inside R7, with R6 met and R11 mass at risk (STR-CAL-001 v0.4).
- TRL 4 work that these decisions point to (measuring efficiency, soil tests that would settle R2, confirming the valve capacity on a bought unit) is decided but on hold, because TRL 4 is on hold by Amish's instruction.
- Cross-repo actions: none.